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Faculty of Medicine
First year of medical studies

Amino acids

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Course Outline:

- I. Structure and Nomenclature
- II. Biological Importance
- III. Classification
- IV. Properties of Amino Acids
- V. Methods for Studying Amino Acids

Learning Objectives:

- Be able to define an amino acid and understand its structure.
- Know the main physicochemical properties.
- Understand the different stages of ionization of an amino acid.

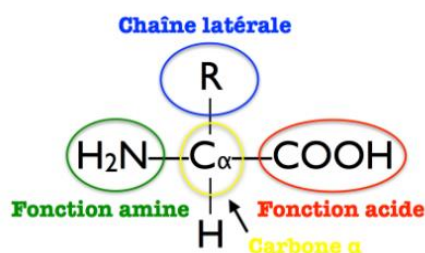
Introduction

Amino acids (AAs) are organic molecules that contain a carboxyl group (-COOH) and an amino group (-NH₂) attached to the same carbon (C_α). They are the building blocks of proteins. More than 300 amino acids have been identified.

I. Structure of Amino Acids

Each amino acid is composed of a central (alpha) carbon bonded to:

- A carboxyl group (-COOH)
- An amino group (-NH₂)
- A hydrogen atom
- A side chain (or R group) that determines the characteristics of the amino acid.



There are only 20 natural amino acids.

I.1. Nomenclature

Amino acids are distinguished from one another by their side chains, which theoretically allows for the creation of an infinite number of amino acids. However, in humans and many other species, only twenty different amino acids are incorporated into proteins during the translation process. To facilitate their identification, an international code using one or three letters is employed to designate each of these twenty amino acids. The table below summarizes the one-letter and three-letter codes for all 20 amino acids.

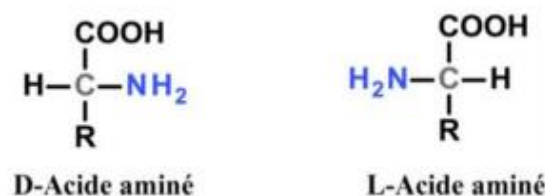
Les 20 acides aminés

Acide glutamique	Glu	E	Leucine	Leu	L
Acide aspartique	Asp	D	Lysine	Lys	K
Alanine	Ala	A	Méthionine	Met	M
Arginine	Arg	R	Phénylalanine	Phe	F
Asparagine	Asn	N	Proline	Pro	P
Cystéine	Cys	C	Sérine	Ser	S
Glutamine	Gln	Q	Thréonine	Thr	T
Glycine	Gly	G	Tryptophane	Trp	W
Histidine	His	H	Tyrosine	Tyr	Y
Isoleucine	Ile	I	Valine	Val	V

I.2. D and L Series

The alpha carbon, which is bonded to four different groups, is asymmetric. Two possible isomers exist: one belonging to the D series and the other to the L series.

- **D series:** NH₂ on the right
- **L series:** NH₂ on the left



Most natural amino acids found in living molecules belong to the L series.

I. Biological Importance

Amino acids play a crucial role in the structure and function of proteins. They are also involved in numerous metabolic pathways and can serve as precursors for other biomolecules.

The biological roles of amino acids are varied:

1. **Protein synthesis:** The primary function of amino acids is to be incorporated into proteins (or peptides) through a series of complex mechanisms known as “protein synthesis.”
2. **Conversion into carbohydrates and lipids:** When the diet provides an excess of amino acids, they lose their amino function and are converted into carbohydrates and lipids, which are then stored.
3. **Biosynthetic precursors:** Some amino acids serve as precursors for substances of interest, either as metabolic intermediates or hormones, such as the synthesis of catecholamines (adrenaline, noradrenaline, and dopamine) from tyrosine.

II.1. Origin of Amino Acids

Amino acids are synthesized by animals, microorganisms, and plants. Their sources are:

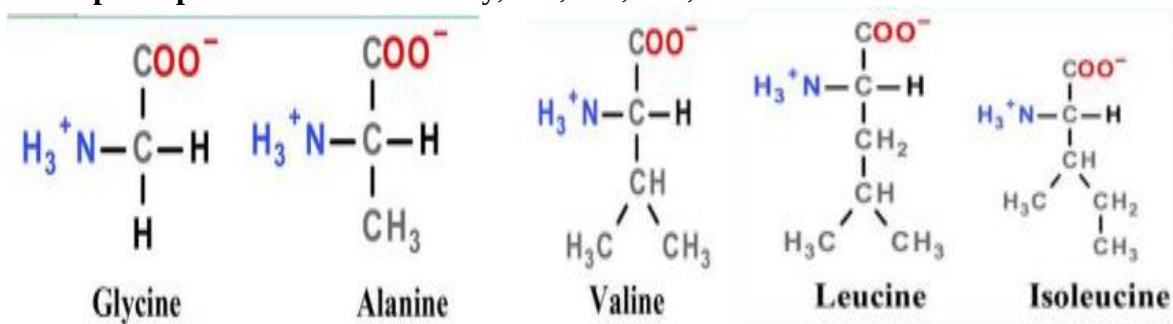
- **Exogenous:** dietary intake
- **Endogenous:** protein catabolism

III. Classification of the 20 natural amino acids

Several classification systems have been proposed, all based on the nature of the R group or specific side chain of amino acids.

III.1. Classification based on the chemical groups present in the R side chain :

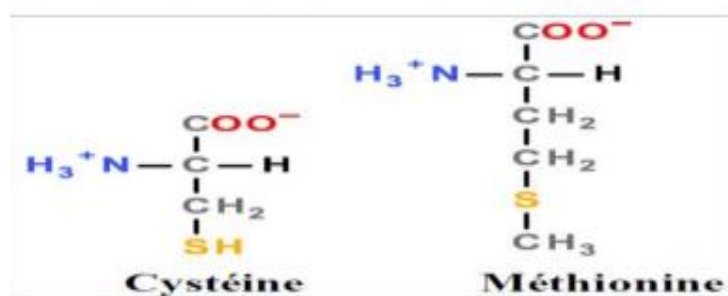
a- Simple aliphatic amino acids: Gly, Ala, Val, Leu, Ile.



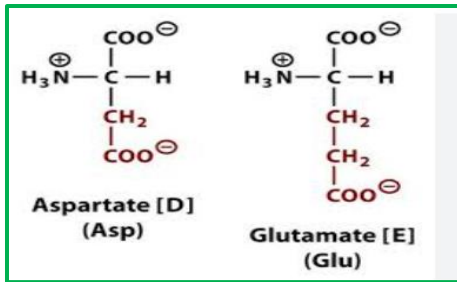
b- Hydroxylated aliphatic amino acids : Ser, Thr.



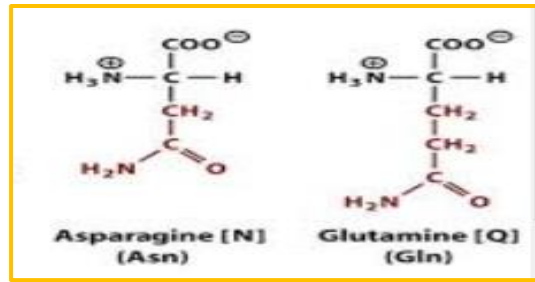
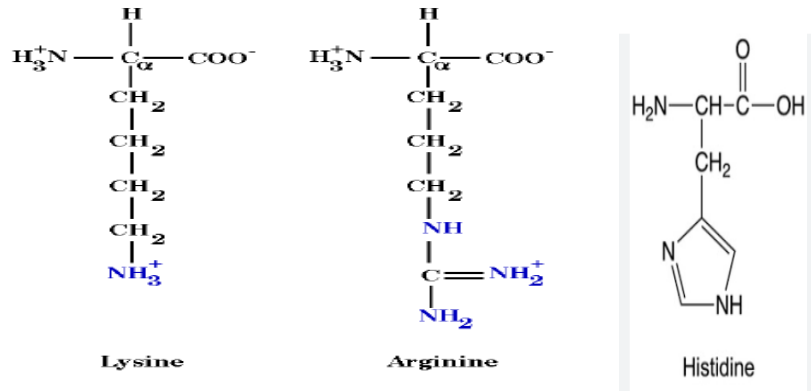
c- Sulfur-containing aliphatic amino acids: Cys, Met.



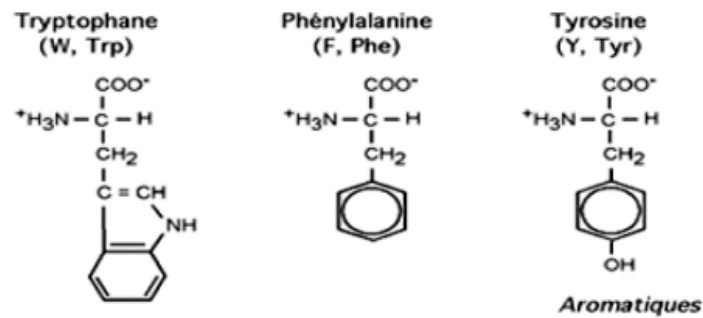
d- Aliphatic dicarboxylic amino acids and their amides: Glu, Asp, Gln, Asn.

**AA diacides**

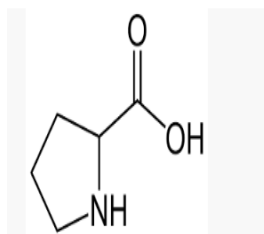
e- Aliphatic diamino acids: Lys, Arg, His.

**Leurs amides**

f- Aromatic amino acids: Phe, Tyr, Trp.

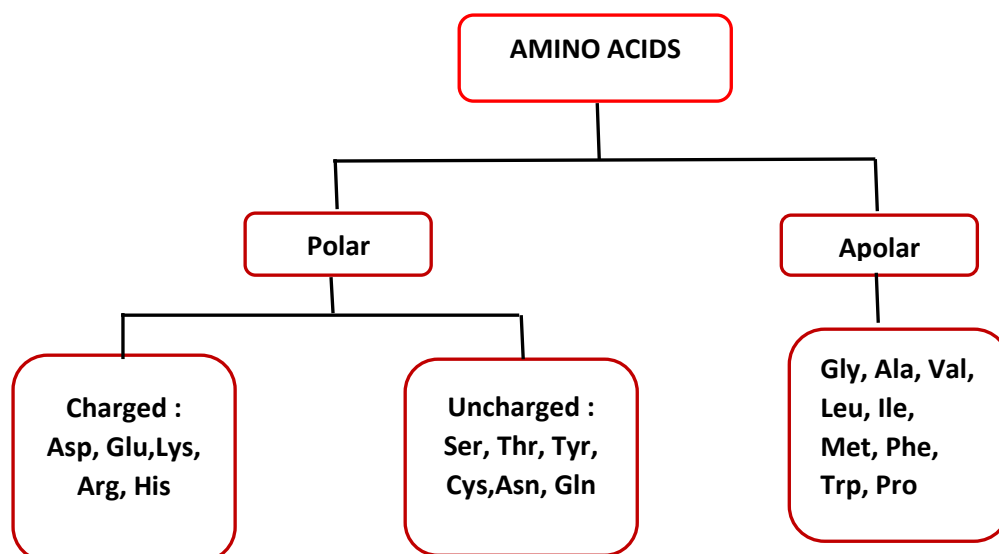


g- Heterocyclic amino acid: Pro.



Proline

III.2. Classification based on the polarity or charge of the R group



III.3. Based on essentiality:

1. **Essential amino acids:** Those that the body cannot synthesize and must obtain from the diet, such as **isoleucine, leucine, methionine, phenylalanine, lysine, threonine, tryptophan, and valine.**
2. **Non-essential amino acids:** Those that the body can synthesize, such as **glycine, alanine, cysteine, serine, glutamate, aspartate, arginine, histidine, tyrosine, proline, asparagine, and glutamine.**
3. **Semi-essential (conditionally essential) amino acids:** Some non-essential amino acids become essential under certain conditions:
 - **Histidine** and **arginine** are essential in **children** (growth) and in **pregnant women**.
 - **Arginine** is essential in **infants**.

VI. Properties of Amino Acids

VI.1. Physical Properties

a. Solubility

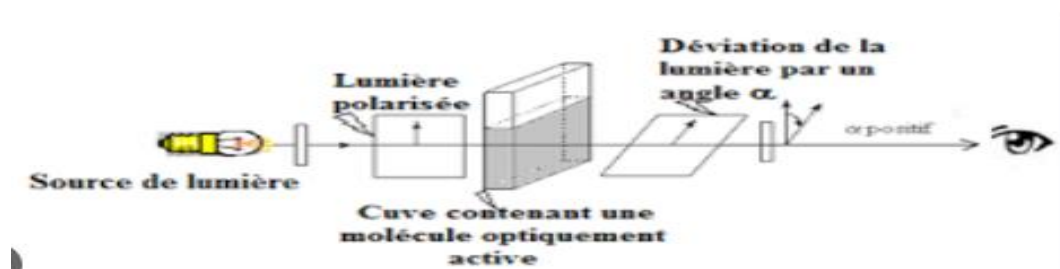
Amino acids are generally soluble in water. Their solubility in this solvent is influenced by the side chain and increases:

- when the number of carbons in the R group decreases,
- if R contains polar functional groups such as COOH or NH₂, or hydrophilic groups such as OH.

Conversely, amino acids have low solubility in alcohol.

b. Optical activity

All amino acids possess an asymmetric carbon (a carbon atom bonded to four different substituents) except glycine, which does not. Therefore, amino acids exhibit **optical activity**, meaning they can rotate plane-polarized light. When these amino acids are placed in a beam of plane-polarized light, they cause the plane of polarization to rotate.



- When the rotation occurs **clockwise**, the molecule is called **dextrorotatory (+)**.
- When the rotation occurs **counterclockwise**, the molecule is called **levorotatory (-)**.
 - The **specific rotation** of a substance in solution is expressed by **Biot's formula**:

$$[\alpha]_D^{20} = \frac{\alpha}{c \times l}$$

Where:

α: Angle of deviation of the polarized light (in degrees)

c: Concentration of the solution (in g/mL)

l: Length of the tube containing the solution (in dm)

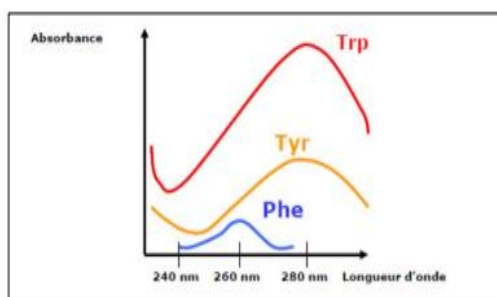
c. Spectral Properties :

Amino acids absorb in the **far UV** region, at wavelengths < 220 nm.

Cysteine absorbs at 240 nm.

Phenylalanine (Phe) absorbs at 260 nm.

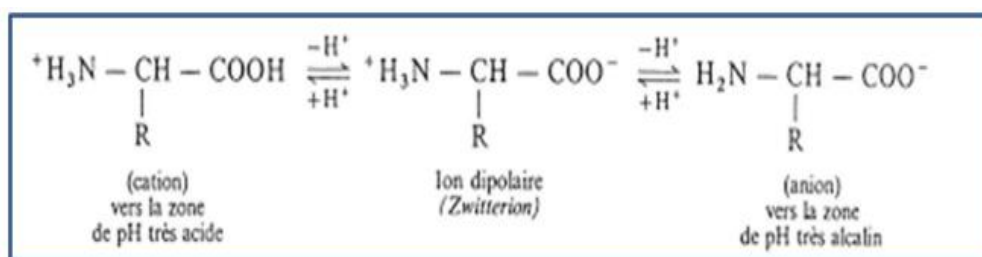
Tyrosine (Tyr) and **Tryptophan (Trp)** absorb at 280 nm, which allows **quantitative evaluation of proteins** using a spectrophotometer at the 280 nm wavelength.



VI.2. Chemical Properties

Amphoteric Character

Amino acids are **amphoteric molecules**, meaning they can act as both acids and bases. Due to the presence of at least one acidic group (COOH) and one amino group (NH₂), an amino acid can exist in **three forms** depending on the pH of the aqueous solution:



At acidic pH (rich in H⁺ protons): The carboxyl group (–COOH) is neutral, while the amino group captures a proton and ionizes into a positively charged cation (–NH₃⁺), giving a total net charge of +1.

At pH near the isoelectric point (pI): The carboxylate group (–COO[–]) is negatively charged, and the ammonium group (–NH₃⁺) is positively charged, making the overall molecule neutral. This forms a zwitterion with a net charge of 0.

At basic pH (low in H⁺ protons): The carboxylate group ionizes to –COO[–], carrying a negative charge after losing a proton, while the amino group (NH₂) remains neutral, resulting in a total net charge of –1.

Remarks

pI: The isoelectric point of a protein, i.e., the pH at which the protein has a net charge of zero (zwitterion form) and does not migrate in an electric field.

For neutral amino acids, pI ranges from 4.8 to 6.3.

For basic amino acids, pI ranges from 7.8 to 10.8.

For acidic amino acids, pI ranges from 2.7 to 3.2.

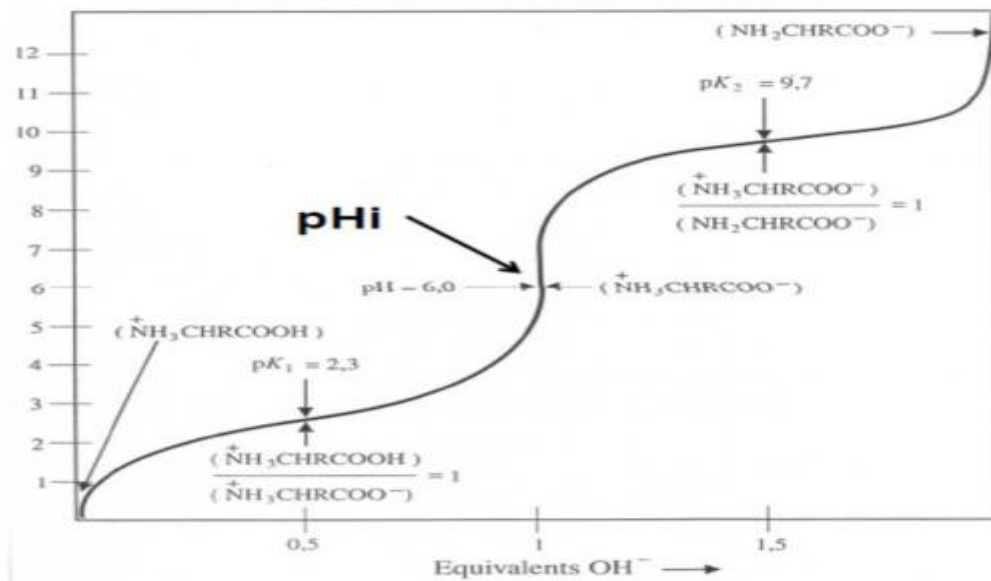
pK: Defines the pK_a and pK_b (or pK₁ and pK₂), which correspond to the pH at which 50% of the acidic or basic groups are ionized.

Thus, the pHi is calculated as follows:

pI=(pK₁+pK₂)/2 For an amino acid whose side chain is not ionizable.

pI=(pK₁+pK₃)/2 For an amino acid whose side chain contains an acidic group.

pI=(pK₂+pK₃)/2 For an amino acid whose side chain contains an amino group.



Courbe de titration d'un acide aminé neutre

To determine the net charge of an amino acid, it is sufficient to compare the pH of the environment with the amino acid's pI (isoelectric point):

pH of the medium $< \text{pH}_i$ $\longrightarrow$ The amino acid will have an overall positive charge

pH of the medium $> \text{pH}_i$ $\longrightarrow$ The amino acid will have an overall negative charge

pH of the medium $= \text{pH}_i$ $\longrightarrow$ The amino acid has an overall neutral charge

V. Methods for studying amino acids

Different methods are used:

Chromatographic methods

Electrophoretic methods

V.1. Chromatographic Methods

Chromatography is a technique that allows the separation of mixtures through an equilibrium between a mobile phase (liquid or gas) and a stationary phase, which is usually solid (column, paper, gel). Several types of chromatography are used:

A. Thin Layer Chromatography (TLC): This is a partition chromatography between a stationary phase, which is a uniform thin layer of gel (hydrophilic), and the mobile phase (eluent), which is the solvent (hydrophobic).

- The analysis takes place in several steps:
- Spotting the sample (of unknown composition) and standards (of known composition) at the extremity of the plate.
- Placing the plate in a closed chamber.
- Migration of amino acids by capillarity, with the most hydrophobic migrating the farthest.

- Visualization of the sample spots using ninhydrin.
- Calculation of retention factors (Rf) and interpretation of the results.

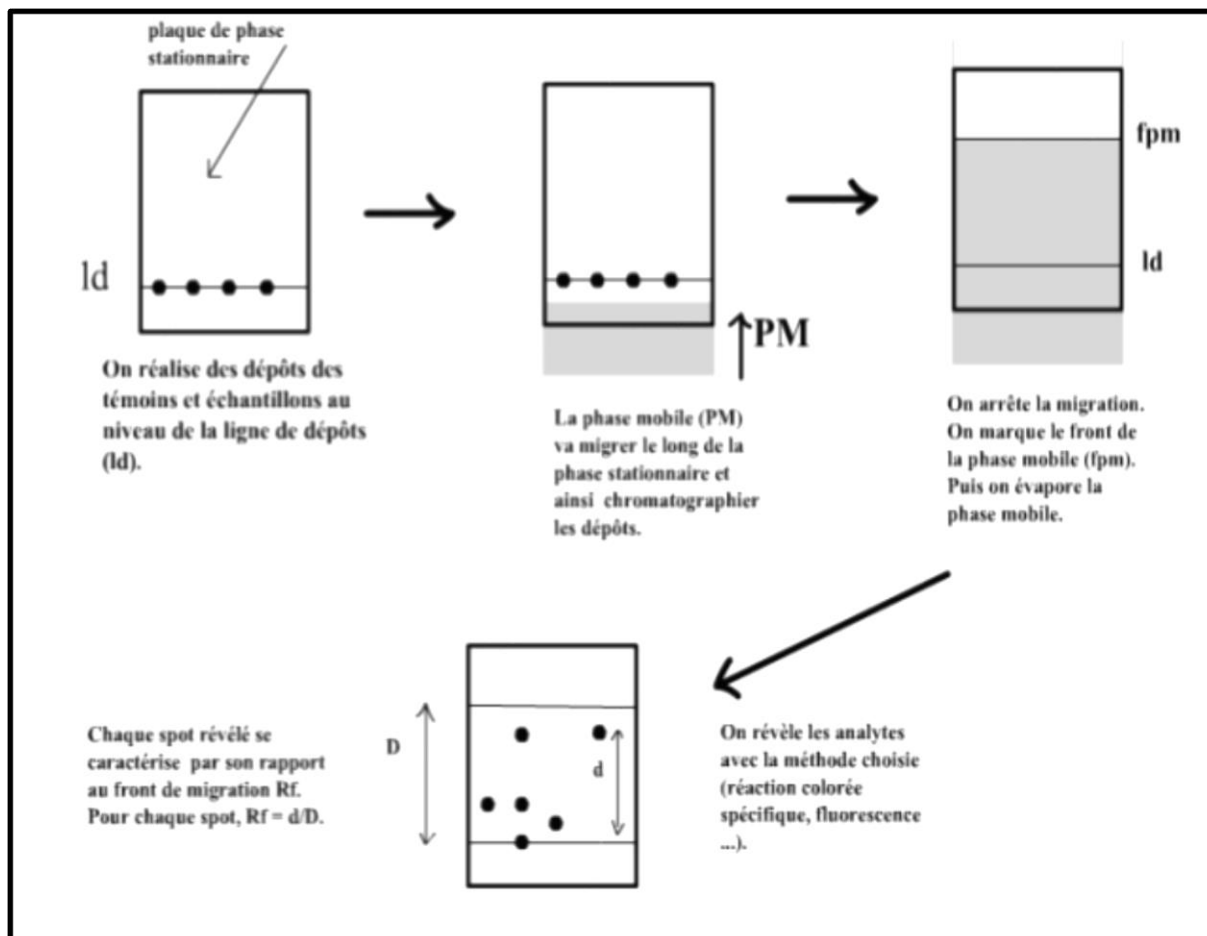


Figure : Steps of Thin-Layer Chromatography (TLC)

$$Rf = \frac{\text{distance parcourue par le composé}}{\text{distance parcourue par l'éluant}}$$

- Rf values range from 0 to 1
 - If the solute is more soluble in the stationary phase, Rf is low.
 - If the solute is more soluble in the mobile phase, Rf approaches 1.

B. Paper Chromatography: This is a partition chromatography between a stationary phase, which is the paper sheet (hydrophilic), and a mobile phase, which is the solvent (hydrophobic).

- It is performed in several steps, which are the same as those used in thin-layer chromatography (TLC).

C. Column Chromatography: The stationary phase is placed in a column, through which the mobile phase passes. The two phases are in intimate contact. The main types of column chromatography are:

- **Gas Chromatography (GC):** Partitioning occurs between a **mobile phase**, which is a chemically inert gas, and a **stationary phase**, represented by the column.
- **Ion-Exchange Chromatography:** In this method, a chromatography column is packed with a **synthetic resin containing charged groups**. There are two classes of ion-exchange resins:
 - **Cation exchangers** (negatively charged)
 - **Anion exchangers** (positively charged)
 The **mobile phase** (eluent, a solution) is poured at the top of the column.
 Amino acids are typically separated using a **cation-exchange column**.
- **High-Performance Liquid Chromatography (HPLC):** The **mobile phase** is driven through the column under **high pressure** using high-pressure pumps.

V.2. Electrophoretic Methods

Principle: The separation of amino acids is based on their differences in mobility within a support subjected to an electric field in a buffered medium.

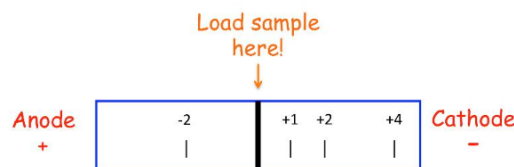


Figure : Electrophoresis

The migration rate of each amino acid depends on two factors:

- The **pH of the buffer solution**.
- The **electric charge of the amino acid**, determined by its **isoelectric point (pI)**.

Migration of amino acids:

- If **pH > pI**: the amino acid is **negatively charged (-)** and migrates toward the **anode** (positive electrode).
- If **pH = pI**: the amino acid exists as a **zwitterion** and **does not migrate**.
- If **pH < pI**: the amino acid is **positively charged (+)** and migrates toward the **cathode** (negative electrode).

Conclusion:

Amino acids are essential molecules that play a central role in protein biology. Their structure, classification, and properties are fundamental for understanding biochemistry and cellular biology.

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